ADEQUATE MODULES ON AN AFFINE SCHEME ARE EQUIVALENT TO ADEQUATE FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof's closing sentence 'We omit the proof that the constructions define mutually inverse equivalences of categories.' is delivered verbatim; it names one specific routine verification (the functoriality check for two displayed constructions) rather than an unstated argument, and the constructions themselves are in the span.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Adequacy of the associated functor of an adequate module. Same chapter, Lemma lemma-adequate-local, invoked at proof lines 1745-1746. Gives one direction of the construction.
- The quasi-coherent module associated to a module, and the presentation of an adequate functor as a kernel. The definition of M^a used in the proof and the definition of adequate functors. Supplies the second construction.

Source reference. Original lemma 06VI: sources/stacks-adequate/pinned-original-source.tex, source lines 1707–1707. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $S = \mathrm{Spec}(A)$ be an affine scheme. The category of adequate $\mathcal{O}$-modules on $(Sch/S)_\tau$ is equivalent to the category of adequate module-valued functors on Alg_A.*

Proof. Given an adequate module $\mathcal{F}$ the functor $F_{\mathcal{F},A}$ is adequate by Lemma 06VI. Given an adequate functor F we choose an exact sequence $0 \to F \to \underline{M} \to \underline{N}$ and we consider the $\mathcal{O}$-module $\mathcal{F} = \mathrm{Ker}(M^a \to N^a)$ where M^a denotes the quasi-coherent $\mathcal{O}$-module on $(Sch/S)_\tau$ associated to the quasi-coherent sheaf $\widetilde{M}$ on S. Note that $F = F_{\mathcal{F},A}$, in particular the module $\mathcal{F}$ is adequate by definition. We omit the proof that the constructions define mutually inverse equivalences of categories. □

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.